

0 6

Two methods that can be used to improve the security of a network are a firewall and MAC address filtering.

0 6 . 1

Describe how a firewall works to keep a computer network secure.

[2 marks]

0 6 . 2

Describe how MAC address filtering works to keep a computer network secure.

[2 marks]

4

Question	Part	Marking guidance	Total marks
05	4	The fact that chicken breast is meat is represented twice/more than once // the category of an ingredient is stored each time the ingredient is used (in a dish); NE. the category field	1 AO2=1

Question	Part	Marking guidance	Total marks
06	1	Analyses/scans network traffic/data // acts as a barrier between Internet and (local area) network; Uses (programmed) rules to determine which traffic/data to allow into network / block from entering network / block from leaving network; A. “software” for “(programmed) rules” Can block traffic/data from/to suspicious hosts/ computers/addresses; Can block traffic/data from/to certain ports/certain types of traffic/data; Max 2	2 AO1=2

Question	Part	Marking guidance	Total marks
06	2	Each (networked) device has unique <u>hardware/physical/MAC</u> address; To connect to a network a device must have an address on the database/allow list (A. whitelist); A. must be a known/registered device	2 AO1=2

0	5
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Pharming and phishing are two social engineering techniques.

0	5	.	1
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Explain what **pharming** is.

[2 marks]

0	5	.	2
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Describe **two** ways a business could minimise the risks from **phishing**.

[2 marks]

4



0 6

Describe why the following are threats to cyber security.

[4 marks]

Misconfigured access rights

Outdated software

4

Turn over for the next question**Turn over ►**

Question	Part	Marking guidance	Total marks
04	3	Faster data transmission; A. fewer collisions occur NE. just “faster” Better security; A. improved privacy NE. safer No reliance on main cable // if a cable breaks only one machine affected/loses connectivity; A. more reliable NE. if one component/part breaks, no others affected R. if one computer breaks, no others affected Easier to locate faults; Max 2	2 AO1=2

Question	Part	Marking guidance	Total marks
04	4	Cheaper to install; A. “cheaper” as BOD NE. easier Uses less cabling; No reliance on switch/hub/central device; Max 1	1 AO1=1

Question	Part	Marking guidance	Total marks
05	1	Redirecting website traffic; (to a) fake website // a website that steals your personal data // another server;	2 AO1=2

Question	Part	Marking guidance	Total marks
05	2	Note: Measures must be ones that a business could take, not that an individual would take. Train employees (on how to spot phishing attempts); Block (external) email // only allow receipt of emails from listed/known email addresses; NE. verify who sent an email Use a mail filter // block email with suspicious content; Limit employee access rights to only the data they need; Use software that checks the URL of a website against a list of known dangerous sites before the website is accessed; A. block websites known to be dangerous // only permit access to websites known to be safe NE. use of firewall Provide a system to report phishing emails; Notify customers that the business will never request login/personal details in an email; Max 2	2 AO1=2

Question	Part	Marking guidance	Total marks
06		If access rights are set-up incorrectly then this could: - allow users access to data that they should not have; A. a hacker/unauthorised/untrusted person could access/change/delete your data A. private / confidential data could be accessed/changed/deleted A. data can be stolen - allow users to edit/delete data <u>that they should only be able to read</u>; - allow external access to data/parts of a network that should be for internal use only; If software is outdated then this could: - mean that there are security flaws/vulnerabilities/weaknesses in the software that are exploitable // updates may fix security flaws; NE. outdated software is more vulnerable - make systems more vulnerable to malware attacks; - make systems more vulnerable to hacking; NE. recent updates may not have been applied Max 3 for misconfigured access rights Max 3 for outdated software Max 4	4 AO1=4

Question	Part	Marking guidance	Total marks
07	1	Unique identifier/field; NE. unique key A. value(s) that uniquely identifies each record	1 AO1=1

Question	Part	Marking guidance	Total marks
07	2	2 marks: A <u>primary key</u> in one table that is <u>present/a field/a key/a value/used in/in</u> another table DPT. “database” for “table” If 2 marks not awarded, then award 1 mark for either: - A field (A. key) that appears in two tables // a shared field - A field (A. key) that links/creates a relationship between/connects two tables A. “more than one” for two	2 AO1=2

Instead of storing files on the network's file server, a business could use cloud storage.

0 9 . 3

State **two** advantages for a business of using cloud storage to store files instead of their network's file server.

[2 marks]

8

1 0

A computer virus is one type of malware.

1 0 . 1

Describe what a computer virus is.

[3 marks]

1 0 . 2

State **one** other type of malware.

[1 mark]

4

Turn over ►



Question	Part	Marking guidance	Total marks
09	3	(Nearly) unlimited storage capacity; A. more storage capacity A. storage capacity can be easily expanded Easy integration with cloud computing services; Makes working from home easier for employees // data accessible from anywhere; Might reduce hardware costs; NE. cheaper Reduced need for in-house technical skills; Maintenance work / back-ups / malware protection done for them // security managed for them; NE. more secure Max 2 R. answers that would also apply to storing files on a central server on the network, eg make it easier to work collaboratively, unless they are expanded to make distinct from storing on a server, eg make it easier to work collaboratively with other businesses	2 AO1=2

Question	Part	Marking guidance	Total marks
10	1	(A program / file / piece of software that): • is attached to another file/program; • executes when the file/program it is attached to is opened / program is executed; • makes copies of itself / spreads / replicates; • (when executed) causes harm to a computer system / other files; Max 3	3 AO1=3

Question	Part	Marking guidance	Total marks
10	2	Trojan; spyware; adware; worm; ransomware; rootkit; Max 1	1 AO1=1